Enhancing Localization Accuracy for Autonomous Mining Vehicles

Project Code: M-21-TYIN-004 **Discipline: Mechatronics Engineering**

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Background: Localization Drops

- Enhancing the localization accuracy of autonomous vehicles used in mining, where traditional GPS signals can be weak or unavailable and similar applications.
- Fusing data from **LiDAR (Light Detection and Ranging)** and **IMU (Inertial Measurement Unit)** sensors, the system can track the vehicle's movement more reliably in subterranean conditions.
- Improved localization supports the development of safer and more efficient autonomous mining operations, especially in complex underground or remote environments.

Objective: Sensor Fusion

- **Sensor fusion** system that integrates LiDAR and IMU data for improved localization and mapping.
- Improve the positioning reliability of autonomous mining vehicles in GPS-denied environments, such as tunnels or remote sites.
- Implement real-time algorithms to process sensor data under challenging and dynamic mining conditions.
- Safer and more efficient autonomous mining operations by reducing reliance on GPS.

Method: Unscented Kalman Filter

UKF-Based Sensor Fusion Localization

- **Unscented Kalman Filter (UKF)** is implemented to fuse LiDAR Adaptive Monte Carlo Localization (AMCL) pose data and IMU orientation data for accurate pose estimation.
- **Robot Operating System (ROS)** robot_localization package was used to build the localization and sensor fusion algorithms.

Simulation: ROS + Gazebo + Rviz

- Simulated mining environment was created in **Gazebo**, where a virtual mining vehicle and sensors were deployed.
- **RViz** was used to visualize trajectories and evaluate performance.

Real-World Testing

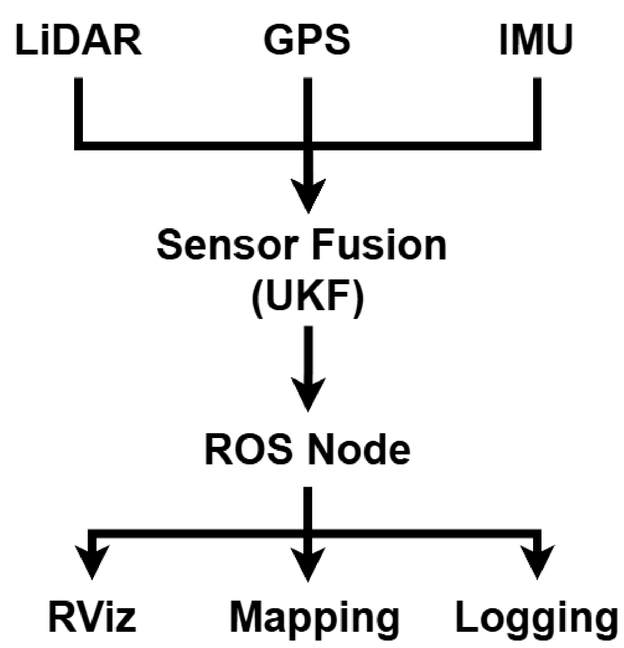
- **Cartographic Hybrid Autonomous Robot for Observation and Navigation (CHARON)** was prototyped with an RPLiDAR A1 and a WT901 IMU for real-world testing.
- ROS system subscribed to sensor data and ran the localization and fusion algorithms onboard.

SLAM: Simultaneous Localization and Mapping

- To verify the accuracy of the UKF-based localization, we used **Cartographer (SLAM system in ROS)** to generate real-time maps.
- Provided a reliable reference to compare against the UKF trajectory.

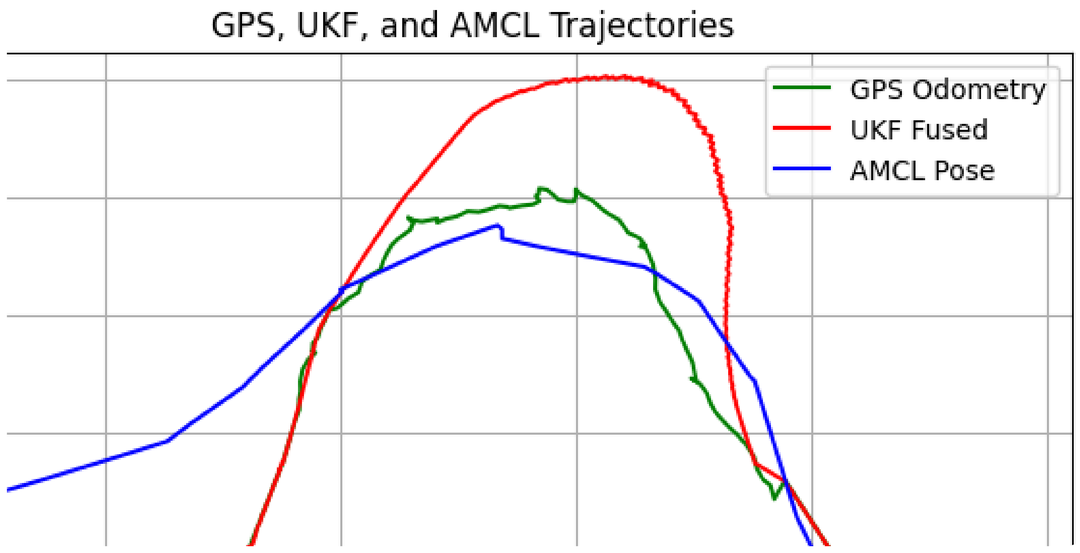


Data Flow

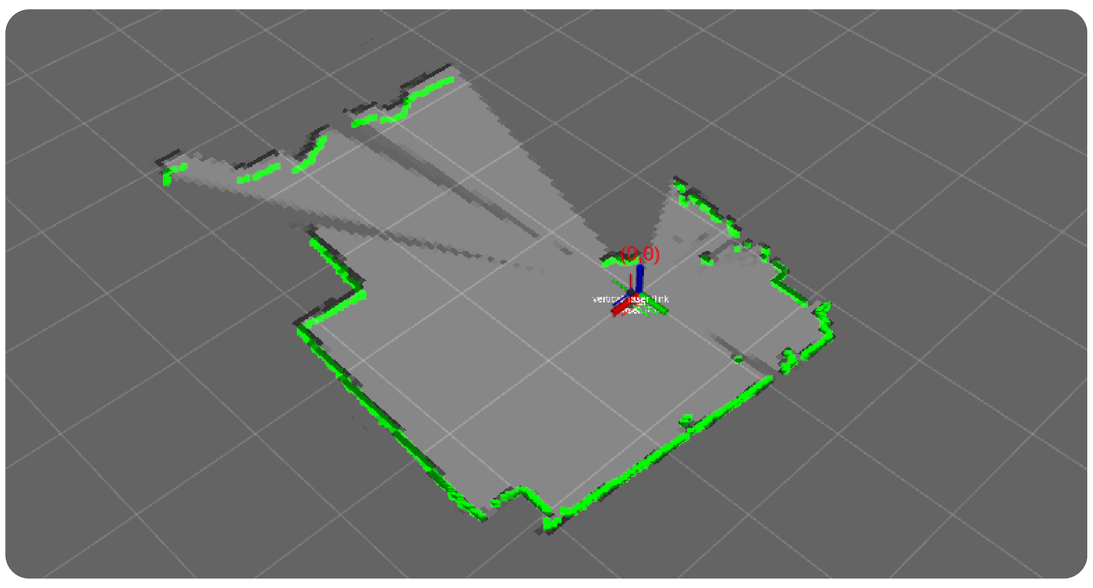


Results and Validation: ROS + UKF + SLAM

- Simulation experiment sensor data shows UKF fused trajectory overlaid with GPS and AMCL trajectory, demonstrating reliable accuracy even in GPS-denied zones.



- Real-world experiments show stable performance of UKF localization with LiDAR and IMU fusion, visualized through Cartographer SLAM mapping.



UKF System States, Vectors & Motion Model

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \theta \\ \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

- The system's state vector shows position, orientation, and velocities.
- UKF uses a constant acceleration motion model.
- UKF package calculates missing state values.

$$\mathbf{x}_{k+1} = \begin{bmatrix} x_k + \dot{x}_k \Delta t \\ y_k + \dot{y}_k \Delta t \\ \theta_k + \dot{\theta}_k \Delta t \\ \dot{x}_k \\ \dot{y}_k \\ \dot{\theta}_k \end{bmatrix}$$

$$\mathbf{z}_{gps} = \begin{bmatrix} x \\ y \end{bmatrix}$$

GPS gives position in x and y

$$\mathbf{z}_{imu} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

IMU gives orientation θ and angular vel θ̇

$$\mathbf{z}_{amcl} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$

LiDAR gives position x, y and orientation θ

Conclusion: Reliable Inertial Localization System

- LiDAR and IMU fusion-based positioning system developed demonstrates high positioning accuracy and robustness in GPS-denied mining environments. Both simulation results and real-world robot testing validate strong stability and reliability.
- Compared to conventional GPS-dependent methods, this sensor fusion approach offers significantly greater continuity and resilience, particularly in areas where GPS signals are weak, intermittent, or completely unavailable. This mitigates the risks associated with reliance on a single sensor modality.
- Current tests have focused on small-scale environments, further evaluation is required in more complex and dynamic mining scenarios to fully establish the system's scalability and operational reliability in real applications.